

SECURE EMS — TECHNICAL MODIFICATIONS REPORT

Summary of Architectural Enhancements & Standalone Applications

Executive Overview

Today's engineering effort focused on modularizing the **Secure Examination Management System (Secure-EMS)** into distinct, dedicated standalone web applications for Central Admin Controllers, Exam Hall Supervisors, and Student Examinees. This architecture enforces strict separation of concerns, improves operational security, and eliminates clutter across interfaces while maintaining end-to-end cryptographic protection.

1. Modular Standalone Web Applications Architecture

Application	Route	Target File	Key Responsibilities & Security Capabilities
Central Admin Terminal	/admin	AdminTerminal.jsx	Question Paper Upload & 2-Stage Fernet Encryption Engine, Key Split Generation (Key A: Admin Token + Key B: Supervisor PIN), Registered Papers Repository Table, & Live Audit Trail.
Supervisor Terminal	/supervisor	SupervisorTerminal.jsx	10s Time-Lock Release Enclave Countdown, Dual-Key Verification Form, Forensic Stamped Watermarking (Center/MAC/IP/Time), Real-Time Student Desk Status Grid (3s Polling), & Session Auto-Lock.
Student Kiosk Terminal	/student	StudentTerminal.jsx	Client-Server Secure Locked Exam Reader Terminal, Active Focus Loss & Tab-Switch Anti-Cheating Alerting, & Automated Heartbeat Status Pings.
Central Dashboard	/	ExamDashboard.jsx	Integrated Hub providing tabbed previews, overall system overview, and initial setup controls.

2. Deep Dive into Newly Implemented Standalone Applications

A. Central Admin Terminal (/admin)

- **2-Stage Encryption Engine:** Uploads raw question paper content and configures a time-lock release delay.
- **Split Authority Key Release:** Generates two-stage authorization credentials: Key A (Admin Controller Token) and Key B (Supervisor Cryptographic PIN).
- **Repository & Live Audit Log:** Displays a real-time table of encrypted papers registered in SQLite backend and live security events.

B. Supervisor Terminal (/supervisor)

- **Time-Lock Release Enclave:** 10-second countdown release timer before authorization input opens.
- **Two-Person Rule Authorization:** Requires simultaneous submission of Admin Key A + Supervisor Key B to reconstruct Fernet key.
- **Forensic Dynamic Watermark:** Stamped background watermark with Center Code, Device MAC, IP address, and timestamp with right-click/copy protection.
- **Real-Time Live Student Monitor Grid:** Auto-polls student statuses every 3 seconds rendering status badges (■ ONLINE, ■ VIOLATION, ■■ DISCONNECTED), violation counters, and ping latencies.

3. Quality Assurance & Automated Testing Verification

Full unit test coverage was established across all standalone components using **Vitest** and **React Testing Library**.

- **AdminTerminal.test.jsx**: Verifies form inputs, API submission, and split-key token rendering.
- **SupervisorTerminal.test.jsx**: Tests time-lock countdown release, dual-key paper decryption, and watermark rendering.
- **ExamDashboard.test.jsx**: Validates auto-logout session timers and navigation.

Result: All 7 automated test suites across 3 test files completed with **100% SUCCESS** and **0 errors**.